The Culture of Business, Finance and the Economy

Class 4 – March 12

Dr. Juan M. del Nido

First semester 2026

Today

Mathematics: from divine perfection to a deal with

Recap

Economics as a kind of culture, with its ways of seeing, of knowing, and its superiority, in part stemming from its mathematical language

Last class: the market as a kind of Order and the place of Economics (and its sister disciplines) in it.

How did Mathematics become what is it today, and also part of economics? Was it always “clear”, “objective”, “perfect”? Was it a usable, or even, a fully developed “tool” or “logic”? Did we discovered it, unearthing it from the secrets of the universe, or did we invent it as a universal means of knowledge?

A class aimed at our “common sense”

LANGUAGE AND THE RISE OF THE ALGORITHM

JEFFREY M. BINDER

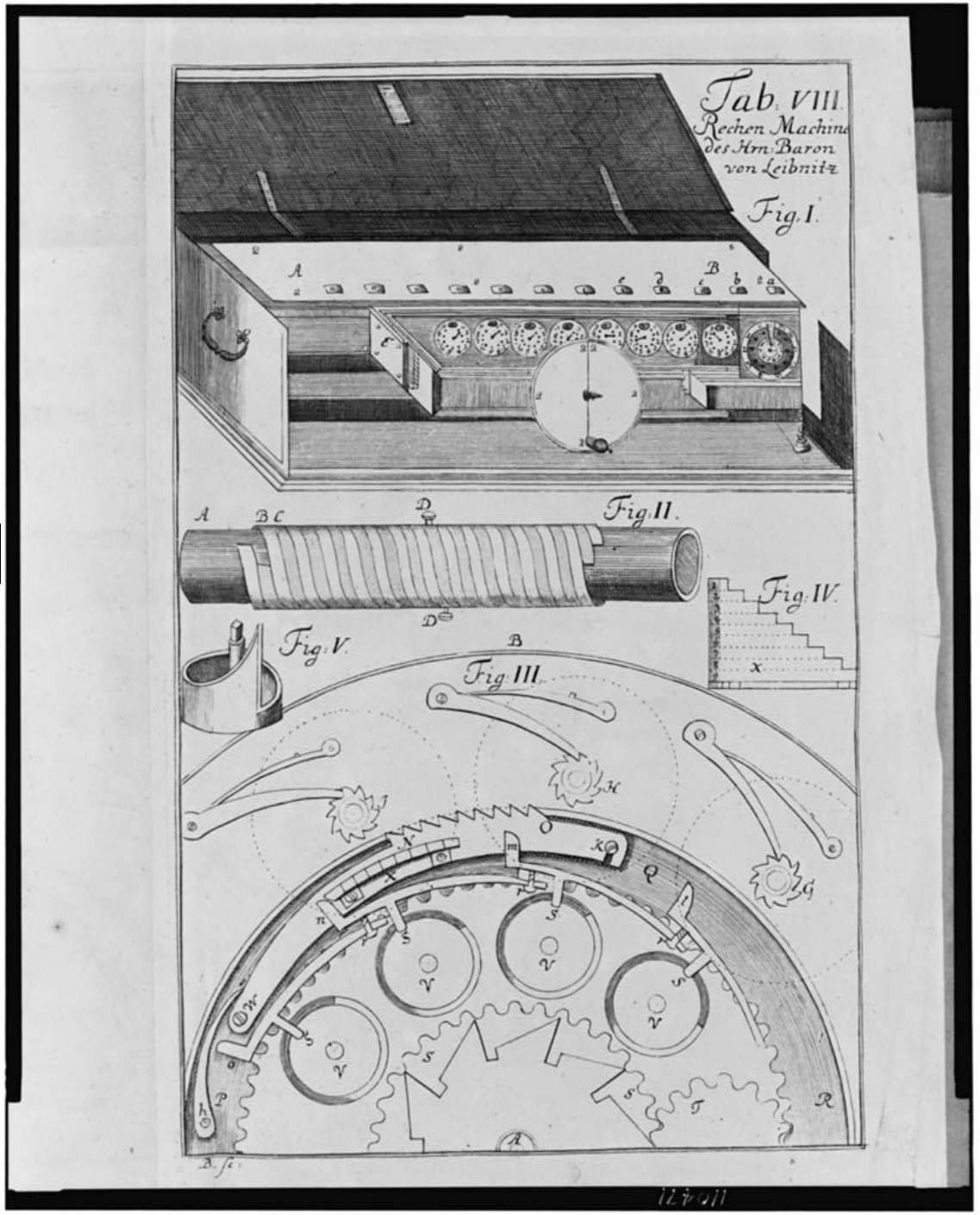
1600s

In 1623, Francis Bacon Vs. words and speech

words “do violence to the understanding” and we need “a new and deeper remedy”

Where was that “new” remedy?

Leibniz’s calculator



Not in “mathematics”

…because what we call mathematics <u>now</u> did not exist <u>back then</u>

The Seven Liberal Arts

"Mantegna Tarot"

15th c.

“the heart of mathematics was geometry. For instance, when Galileo Galilei made his famous statement—in his 1623 book The Assayer—that God wrote the book of the world in the language of mathematics, he was *explicitly referring to geometric diagrams*, not to any sort of symbolic notation”

Descartes

“Arithmetic and Geometry alone are free from any taint of falsity or uncertainty” because they derive knowledge wholly through deduction, not from experience.

Western culture at the time devalued “mechanical” activities, such as calculation.

In Geometry, quantities (“four apples”) and magnitudes (four square meters) are different *kinds* of numbers. So the equation

$$ax^2 + bx + c = 0$$

Did not even make sense, because it “pretends” to add linear and square kinds.

ALOGOS

Since Euclid and the beginnings of Geometry everyone knew of lines whose lengths cannot be expressed as multiples of a common unit

$$\sqrt{2}$$

A proto-algebra existed, but it was considered dirty, plebeian, and not real mathematics, since it was not one of the traditional liberal arts and was perceived to lack rigorous standards of proof

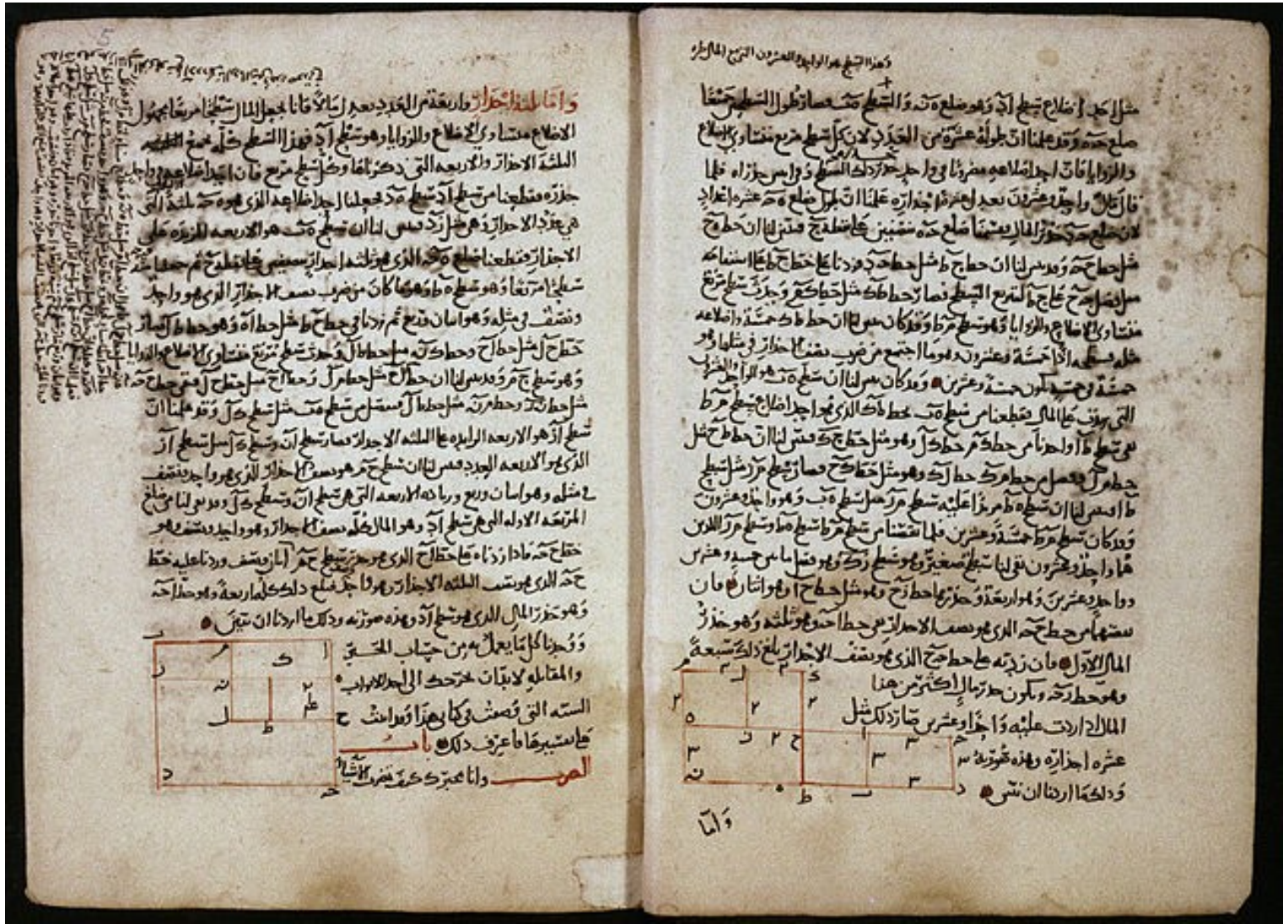
Algebra was a practical, earthly kind of knowledge associated to trivial and merchant, sailing and trade-related calculations

How did this proto-algebra work?

Al-Kwarizmī

House of Wisdom, Bagdad

“Compendious Book on Calculation by Completion and Balancing”, 820 A.D.



How did it work?

“You will add each place to the place that is above it with regard to its own kind, i.e., units to units and tens to tens. When ten has been collected in one of the places, i.e., in the place of the units or tens or in some other place, put a one instead of it and elevate it to a higher place, i.e., if you have ten in the first place which is the place of the units, make a one of it and raise it to the place of the tens and there it will signify ten. But if there remains something from the number that is less than X or the number itself is less than X, leave it in the same place. And if nothing remains, put a circle (i.e., o), so that the place may not be empty”

“Do likewise,” we are instructed, “also in all the places.”

The word algebra is derived from the word *al-jabr* - usually translated as “completion”

Break

(this proto algebra) was primarily the business of “mathematical practitioners”: people who taught mathematics outside of the university system through textbooks, lecturing, and tutoring. The methods they taught had *practical* applications in navigation, trade, and finance, in artisanal trades such as bricklaying and construction, and in military practices such as ballistics. It spread across Europe through merchants, not through philosophers.

algebra was not only a plebeian practice, as opposed to enlightened knowledge. It also brought with it conflicting ideas about what numbers are and how they work.

Working through his proto algebra for merchants and construction sites, al-Khwārizmī also obviously knew about irrational numbers such as $\sqrt{2}$; he instructs the reader:

one must be content with “an approximation, and not the exact truth: for God alone knows what the exact root is.”

ARITHMETICA INTEGRA.

Authore Michaele Stifelio.

Exlibris Pt Doctrine Xpiane Collegij Romani Sti Marie

Norimbergæ ud Iohan. Petr[illegible]um.
Anno Ci[illegible]i M. D. XLIIII.

Cum gratia & priuilegio Cæsareo
atq; Regio ad Sexennium.

Stifel, 1544

"It is disputed of irrational numbers whether they are true numbers, or fictions." In favor of the existence of irrationals is their utility in calculation, on account of which "we are moved to confess that they truly exist, namely by their effects, which we perceive to be real, certain, and constant." On the other hand, when we "try to subject them to numeration, and proportion them by rational numbers, we find that they flee perpetually."

A major shift in conceptions of number

Numbers began to be imagined not through the perfection of arithmetic and geometry, but through the digits themselves... the tools of the commoners who practiced algebra. “there are no absurd, irrational, irregular, inexplicable numbers” Perhaps one cannot write $\sqrt{2}$ as a fraction, but one can write it (in modern notation) as 1.41421356 . . . and continue the expansion as far as one likes

Even further

What about cubic equations, absurd for Euclidean geometry, that yield roots of negative numbers?

the square root of negative one cannot even be approximated by decimal fractions, since no positive or negative number has a negative square. This result could simply be rejected as a sign that the equation has no solution, and for centuries this would remain the typical reaction. Yet the fact that the procedures of algebra could produce such results seemed to imply that the procedures themselves were ill founded, and the problems worsened as mathematicians extended their range.

algebra enters mathematics

François Viète, astronomer

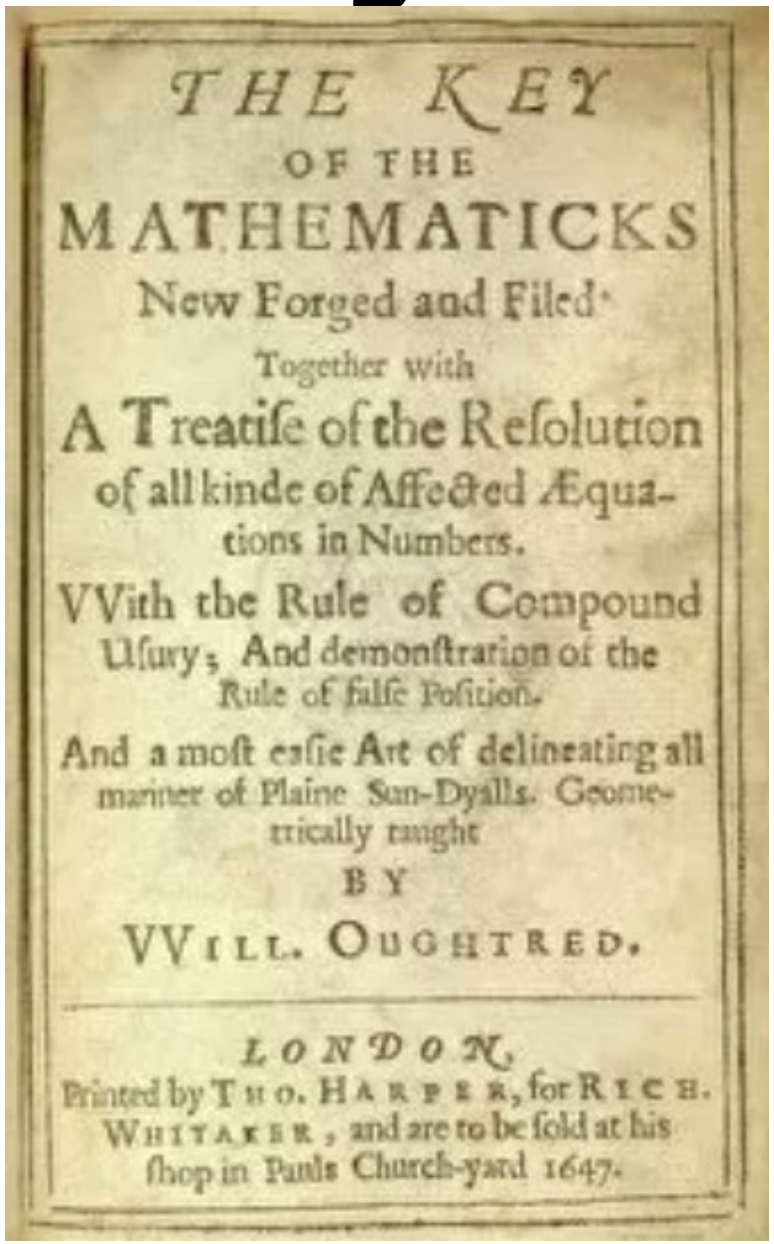
Instead of working on single equations such as 14x + 15 = 71, one could now draw conclusions about general “species” of equation (linear, quadratic, cubic, etc.) dealing with whole unknowns such as

$$Ax + B = C$$

Viète ends the 1591 Introduction to his book with a statement of purpose:

“TO LEAVE NO PROBLEM UNSOLVED”

William Oughtred, 1647



THE KEY
OF THE
MATHEMATICKS
New Forged and Filed:
Together with
A Treatise of the Resolution
of all kinde of Affected Æquations in Numbers.
VVith the Rule of Compound Usury; And demonstration of the Rule of false Position.
And a most easie Art of delineating all manner of Plaine Sun-Dyalls. Geometrically taught
BY
VVILL. OUGHTRED.

LONDON,
Printed by THO. HARPER, for RICH. WHITAKER, and are to be sold at his shop in Pauls Church-yard 1647.

So we can more clearly understand though, let us free propositions and statements from words, and cast them in mathematical symbols plain to the eye.”

Symbols are now a medium free of rhetoric: the naked truth, unadorned, all substance–exactly what Bacon was after.

And so mathematics became thinkable as a means to universal knowledge, to leave, as Viète said, no problem unsolved.

at what cost?

The alphabet is really now superfluous
for in this sign all men can find salvation.

—GOETHE, *Faust, Part II* (trans. Atkins)

THE DEVIL AND Dr. FAUSTUS.

Conclusion